



Mapping recreational marine traffic from Sentinel-2 imagery using YOLO object detection models

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ABSTRACT

Identifying where maritime activities take place, and quantifying their potential impact on marine biodiversity, is important for the sustainable management of marine areas, spatial planning and marine conservation. Detection and monitoring of small vessels, such as pleasure crafts, has been challenging due to limited data availability with adequate temporal and spatial resolution. Here, we develop an analysis framework to detect and quantify small vessels, using openly available Sentinel-2 optical imagery, and the YOLO-family object detection models. We also show how existing spatial datasets can be used to improve the quality of the model output.

We chose five distinct marine areas along the Finnish coast in the northern Baltic Sea and manually annotated 8,768 vessels for training and validating object detection models. The top-performing model, based on F1-score, achieved a test set F1-score of 0.863 and an mAP50 of 0.888. This model was then used to quantify pleasure crafts in the southern part of Finland from May to September 2022, using all available Sentinel-2 images with sufficient image quality. Detected recreational traffic was primarily concentrated on boating lanes and close to marinas, with weekend traffic averaging approximately twice the volume of weekday traffic.

The developed approach can be used to identify areas with high boat traffic, thus supporting sustainable spatial planning, ecological impact avoidance, and serving as an indicator of recreational popularity of marine areas. Additionally, it provides a basis for assessing the potential impacts of pleasure crafts on marine biodiversity. By leveraging openly available satellite images with frequent revisits, the approach offers broad geographical coverage and high spatial accuracy, making it scalable for estimating boat traffic across large areas, even globally.

1. Introduction

Human activities are a major source of pressure in marine environments, contributing to loss of biodiversity and degradation of important habitats (e.g., Dailianis et al., 2018; Virtanen et al., 2024). Among these activities, recreational boating may locally exert considerable pressure on marine ecosystems, especially threatening habitats formed by submerged aquatic vegetation (SAV). Smaller boats, with smaller draughts, can enter shallow coastal areas, where aquatic vegetation typically thrives. SAV creates ecologically valuable habitats, providing shelter for various fish and invertebrate species (Kraufvelin et al., 2018; Santos et al., 2022), contributing to carbon storage (Duarte et al., 2010) and water clarity (Austin et al., 2017), among other high-value ecosystem services (Waycott et al., 2009). Boating may increase

water turbidity and sedimentation, cause habitat loss and the release of harmful substances from sediments through anchoring (reviewed by Sagerman et al., 2020). Furthermore, potential grey water discharges may exacerbate the impacts of recreational boating, particularly in shallow bays where water exchange is limited. Thus, spatio-temporal information on recreational boating is essential for the management and conservation of marine areas but is currently lacking.

While large, commercial vessels are required to use an Automatic Identification System (AIS) to broadcast their location, smaller pleasure crafts do not have an AIS requirement. Moreover, AIS does not provide complete information about marine traffic, as it has been estimated that up to three-quarters of commercial marine traffic operates with-

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out it (Paolo et al., 2024). Information on the amount of pleasure craft movement is typically acquired through questionnaires or visitor surveys, logging the visits to marinas and harbours, or with visual observations and field work (Gray et al., 2011; McCole and Joppe, 2014; Venturini et al., 2021). While these methods provide accurate information, they have limited spatial and temporal coverage, also with higher costs of acquiring data (Ruciński et al., 2023).

Previous research on vessel detection has mainly concentrated on larger ships (reviewed by Kanjir et al., 2018), but there are few studies on small boat detection using visual coastal observations (Hao and Nabe-Nielsen, 2023), camera-based monitoring (Hartill et al., 2020; Signaroli et al., 2024) and marine radars (Cope et al., 2023). Options for monitoring maritime traffic have increased, along with the development of satellite sensors, with solutions ranging from synthetic aperture radar (SAR) to optical satellite imagery. For instance, several marine vessel benchmark datasets exist which use SAR data (cf. SSDD (Zhang et al., 2021); LS-SSDD-v1.0 (Zhang et al., 2020); SAR-Ship-Dataset (Wang et al., 2019); and HRSID (Wei et al., 2020)), or optical satellites (cf. Airbus Ship Detection Challenge dataset (Kaggle, 2018) and S2-SHIPS (Ciocarlan and Stoian, 2021)). All imagery sources have their advantages and disadvantages. With SAR imagery, it is possible to detect objects even on cloudy days, but image processing and human interpretation is more challenging. High-resolution, open-access optical imagery (e.g., Copernicus Sentinel 2) is easier to interpret for humans than SAR, but clouds prevent observations, and the smallest stationary boats cannot be detected, as they are significantly smaller than the spatial resolution of the data. Very high-resolution (VHR) images (1 metre or less) are available on commercial basis, both as optical and SAR data, and even the smallest boats can be captured using them (e.g. Büyükanber et al., 2023). However, VHR imagery is expensive, limiting the possibilities for monitoring boat traffic with sufficient spatial and temporal coverage. Thus, VHR imagery is impractical for monitoring large areas, or over longer time periods. Nevertheless, no matter which data source is used, the previously published studies about existing methodology is mainly focused on commercial marine traffic, such as fishing fleets and cargo ships instead of recreational boating (Kanjir et al., 2018).

During the recent decade, deep learning (DL) methods have proliferated and taken over the tasks of computer vision, as they have proven to be effective in various domains, from traditional photographs to remote sensing data. Convolutional neural networks (CNNs) have been especially widely adapted to various use cases in remote sensing (Ma et al., 2019; Yuan et al., 2020). They have also proved to be successful in marine vessel detection. For instance, optical VHR-imagery has been used as a data source by Patel et al. (2022), Wen et al. (2023), Büyükanber et al. (2023) and Kong et al. (2023), whereas Xu et al. (2022), Prete et al. (2023) and Ciocarlan and Stoian (2021) used data with around 10-metre resolution. Data sources and results of these studies are collated in Table 1. Direct comparison between these studies is not straightforward due to differences in data sources and processing steps, but they show that state-of-the-art DL solutions are viable for monitoring marine traffic.

In this study, we present a workflow for using object detection models from YOLO family to detect and monitor maritime traffic over broad geographical areas and long time periods. We use openly and freely available Sentinel-2 optical imagery, chosen for its high revisit time and sufficient spatial resolution for our purposes. For the reference data, we manually annotated over 8700 marine vessels from 18 Sentinel-2 products, covering a representative set of Finnish coastal areas. We also demonstrate how the approach can be used to quantify the hotspots for marine traffic, which has applications in the management and conservation of shallow coastal areas.

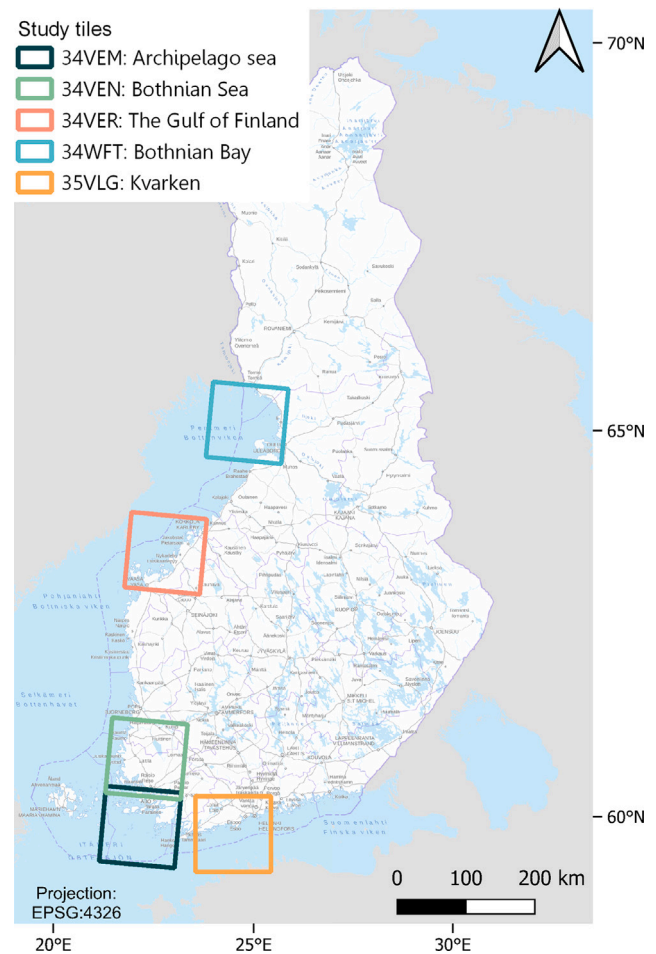


Fig. 1. Locations of the study areas in the Finnish coast. Background map: The National Land Survey Finland.

2. Materials

2.1. Study areas

To obtain comprehensive data from the Finnish coast, we selected five separate Sentinel-2 tiles as training and testing locations. These tiles cover the Archipelago Sea (34VEM), the Bothnian sea (34VEN), the Gulf of Finland (35VLG), Kvarken (34VER) and the Bothnian Bay (34WFT) (Fig. 1 and Table 2). The areas were chosen to represent a range of varying coastal gradients with respect to morphology and ocean colour. Optically, these areas represent mostly absorption-dominated Case II waters, with a high absorption of Coloured Dissolved Organic Matter (a_{CDOM}), (e.g., Attila et al., 2013, 2018). High a_{CDOM} makes the waters darker, as it especially absorbs on the bandwidths of blue light (around 430–500 nm). Thus, the reflected signal from the water is very low (e.g. Ylöstalo et al., 2016; Kowalczyk et al., 2005, 2006). While the Archipelago Sea and Kvarken host a high number of islands, skerries and sheltered bays, the coastal areas of the Gulf of Finland, Bothnian sea and Bothnian Bay are more exposed. The Archipelago Sea is heavily burdened with eutrophication and high water turbidity, due to the transport of suspended sediments and particulate organic matter from rivers (Korpinen et al., 2018). Recreational popularity of the areas also differs, as boating activities mainly concentrate in the southern Finnish coast, especially in the Archipelago Sea (Strandell et al., 2020; Paulus et al., 2024).

Table 1
Summary of previous studies of marine vessel detection.

Authors	Satellite	Resolution	Methods	Metrics	Results
Patel et al. (2022)	SPOT6/7 Pléiades Planetscope	0.6–6 m	YOLOv3	F1-Score	0.56–0.61
			YOLOv4	Precision	0.67–0.71
			YOLOv5	Recall	0.44–0.63
				mAP	0.49–0.65
Wen et al. (2023)	Spot6/7 Pléiades	0.6–6 m	MS-SSD	F1-Score	0.66
				Precision	0.57
				Recall	0.81
				mAP	0.71
Xu et al. (2022)	GF-1	8 m	LMO-YOLO	Precision	0.93–0.95
		16 m		Recall	0.86–0.91
				mAP20	0.88–0.92
Prete et al. (2023)	Sentinel-2	10 m	VFNet	mAP	0.24
			Faster-RCNN	mAP50	0.68
			YOLOX		
			RTMDET		
			Mask-RCNN		
			CenterNet		
			FoveaBox SSDLite		
Ciocarlan and Stoian (2021)	Sentinel-2	10 m	U-Net	F1-score	0.53
				Precision	0.47
				Recall	0.77
Büyükkıran et al. (2023)	JL-1 GF-2	0.71–3 m	YOLOv5	mAP	0.81
			YOLOX	mAP50	0.96
			YOLOv7		
			YOLOv8		
Kong et al. (2023)	GF-3	1 m	YOLOx-Tiny	Precision	0.94–0.96
				mAP	0.87–0.96

2.2. Satellite data

Currently, the Sentinel-2 mission consist of two polar-orbiting satellites (Sentinel-2B and Sentinel-2C), equipped with a 13-band Multi-Spectral Imager (MSI) with a swath width of 290 km, and one additional satellite (Sentinel-2A) complementing the two polar-orbiting satellites. However, the data used in this study were collected prior to the launch of Sentinel-2C, when Sentinel-2A and Sentinel-2B had polar-orbiting positions. Each satellite has a revisit time of ten days at the equator, which can provide a new datatake in two or three days in mid-latitudes. The data are available in different processing levels: Level-1C (L1C) products are processed to TOA reflectance and are resampled with a constant Ground Sampling Distance (GSD) of 10, 20 or 60 m, depending on the spectral band. The Level-2A (L2A) products contain Bottom-Of-Atmosphere (BOA) reflectance derived from the L1C products. The L1C and L2A products are distributed as 100 km² orthoimages, also called tiles. These images can either be fully or partially covered by image data, depending on the edges of the swaths (European Space Agency, 2015).

For each location, we selected multiple separate acquisitions to capture scenes with varying conditions during the summer, including algae blooms and variations in water colour (Table 2). We downloaded the raw products from the Copernicus Data Space Ecosystem as L1C products. We opted for L1C products, instead of L2A products, because the Sen2Cor algorithm, which is used to produce L2A products, often has challenges in dark water areas, resulting in image artefacts or incorrect reflectance values. The L1C products were converted into RGB mosaics using a colour-correction method used for visualising satellite images in the Finnish Environment Institute's open EO service, Tarkka (<https://tarkka.syke.fi>). The Finnish coastal waters are, for the most part, particularly dark waters, due to the high absorption of CDOM (Coloured Dissolved Organic Matter) and the low amounts of light, especially in higher latitudes. The true-colour image colour-correction method is based on a tone-mapping process, which compresses the high dynamic range of the original digital numbers (DN) into a visually pleasing image and maintains the highest contrast regions within the darker part of the spectrum to provide more visual resolution for dark

water bodies, which helps with the detection of small targets, at least for human experts. The tone-mapping tables are band specific (B4, B3, B2) and are generated using manual editing, grey-level matching and highlight compression techniques. Technically the tone-mapping is implemented as channel-specific look up tables (LUTs) between Sentinel-2 L1C digital numbers and 8-bit RGB-values.

In contrast, the true-colour image available in Level-1C products, is formulated to provide an overall satisfying colour image throughout waters globally. It consists of linearly rescaled bands B02, B03 and B04 so that the digital numbers between 1 and 3558 are rescaled to 1–255. In this study, we compared how the used conversion method affects the detection of marine vessels. If, for some reason, a TCI was not available in Sentinel-2 products, we constructed it from the R, G and B bands using the same equation. Visual differences of these methods can be seen in Fig. 2. All further visualisations of Sentinel-2 data in this work are presented with only Tarkka colour adjustments.

2.3. Marine vessel annotations

To have sufficient training and validation data for our task, we manually annotated all visible boats and ships from each of the eighteen Sentinel-2 mosaics by drawing bounding boxes around them. These annotations were saved as GeoPackage files, with each layer corresponding to a single RGB mosaic. The annotated data were georeferenced to the corresponding Coordinate Reference System (CRS) with the reference image in order to have orthogonal bounding boxes with the images. As small vessels cover only a few pixels and can be easily confused with different classes of stationary objects, we compared the three images to assess whether the identified objects remained completely stationary in multiple images. The minimum area, which objects needed to cover to be annotated, was 2×2 pixels (20×20 metres or 400 m²). While the minimum size of our annotations was considerably larger than most pleasure crafts, these vessels were in general considerably brighter than their surroundings, and were thus clearly identifiable, even though they were smaller than the spatial resolution of the Sentinel data, as shown in Fig. 3. Also, in some cases

Table 2
Study areas and the Sentinel-2 products used.

Location	Product name
Archipelago Sea	S2A_MSIL1C_20220515T100031_N0400_R122_T34VEM_20220515T120450
	S2B_MSIL1C_20220619T100029_N0400_R122_T34VEM_20220619T104419
	S2A_MSIL1C_20220721T095041_N0400_R079_T34VEM_20220721T115325
	S2A_MSIL1C_20220813T095601_N0400_R122_T34VEM_20220813T120233
The Gulf of Finland	S2B_MSIL1C_20220606T095029_N0400_R079_T35VLG_20220606T105944
	S2B_MSIL1C_20220626T095039_N0400_R079_T35VLG_20220626T104321
	S2B_MSIL1C_20220703T094039_N0400_R036_T35VLG_20220703T103953
	S2A_MSIL1C_20220721T095041_N0400_R079_T35VLG_20220721T115325
Bothnian Bay	S2A_MSIL1C_20220627T100611_N0400_R022_T34WFT_20220627T134958
	S2B_MSIL1C_20220712T100559_N0400_R022_T34WFT_20220712T121613
	S2B_MSIL1C_20220828T095549_N0400_R122_T34WFT_20220828T104748
Bothnian sea	S2B_MSIL1C_20210714T100029_N0301_R122_T34VEN_20210714T121056
	S2B_MSIL1C_20220619T100029_N0400_R122_T34VEN_20220619T104419
	S2A_MSIL1C_20220624T100041_N0400_R122_T34VEN_20220624T120211
	S2A_MSIL1C_20220813T095601_N0400_R122_T34VEN_20220813T120233
Kvarken	S2A_MSIL1C_20220617T100611_N0400_R022_T34VER_20220617T135008
	S2B_MSIL1C_20220712T100559_N0400_R022_T34VER_20220712T121613
	S2A_MSIL1C_20220826T100611_N0400_R022_T34VER_20220826T135136

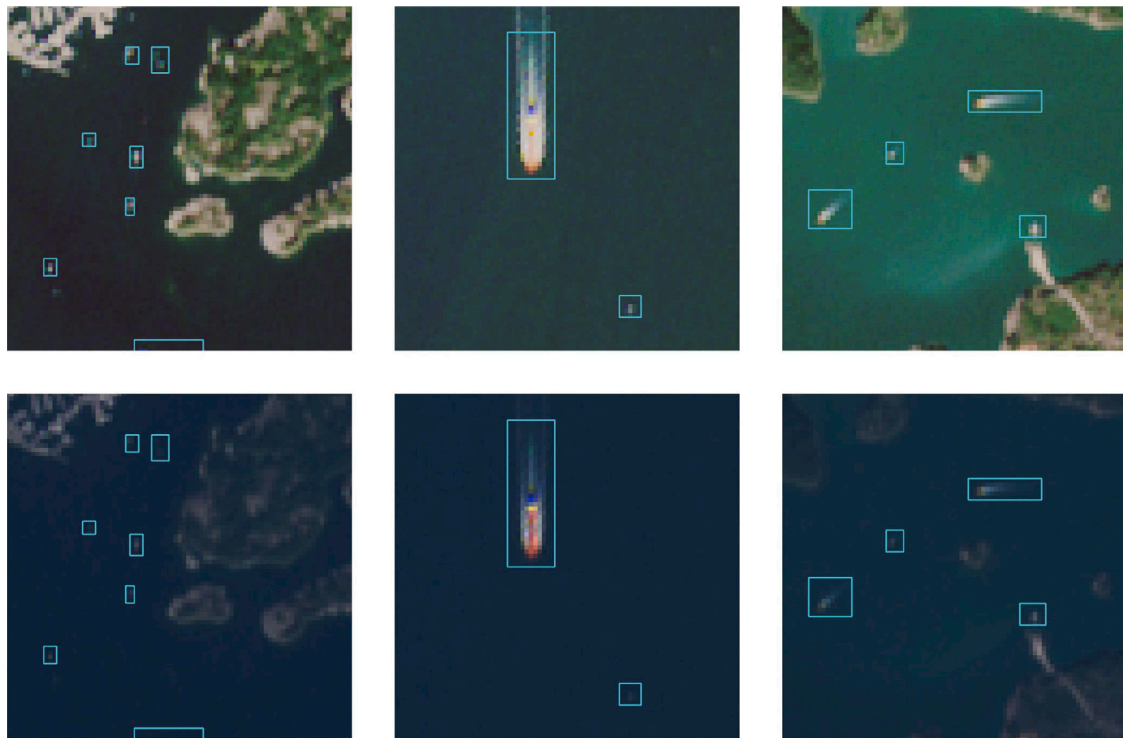


Fig. 2. Visual comparisons of Tarkka colour adjustment (top row) and L1C-TCI images (bottom row). Boxes show examples on the annotated vessels.

the bounding boxes contained parts of the wakes of the moving vessels. This was done in order to make the detection of moving objects easier.

In total, we annotated 8768 marine vessels from these images. The number of annotations in each image ranged between 66 and 1518, with the majority from tiles 34VEM and 35VLG. The dataset does not contain vessels docked at marinas or moored to the shore, as these mix with their surroundings and are thus difficult to detect. The average area of one bounding box was around 5350 m² (53.5 pixels), but over half of the annotations were smaller than 2343 m² (23.4 pixels). Since most annotations also covered wakes of the vessels, the bounding boxes were considerably larger than a typical vessel. There were a few annotations larger than 100,000 m², identified either as large cruise or cargo ships travelling diagonally.

We set two tiles, 34VEN and 34VER, aside to be used as test data. Tiles 34VEM and 34VEN have an overlapping area of 9.5×100 km; therefore, the overlapping area was cut from 34VER, leaving the area available for the training set. Next, as the full-sized satellite images are too large for the object detection models to process as a whole, we split tiles 34VEM, 35VLG and 34WFT into $3 \times 320 \times 320$ pixel chips so that all chips that contained only nodata were discarded. These data were then spatially split into training and validation sets so that 20% of the total area of the tile was used for validating the models during training. All in all, this resulted in 12,193 chips (1975 chips with annotations, 10,218 background chips) used for training and 3141 chips (438 chips with annotations, 2703 background chips) for validation.

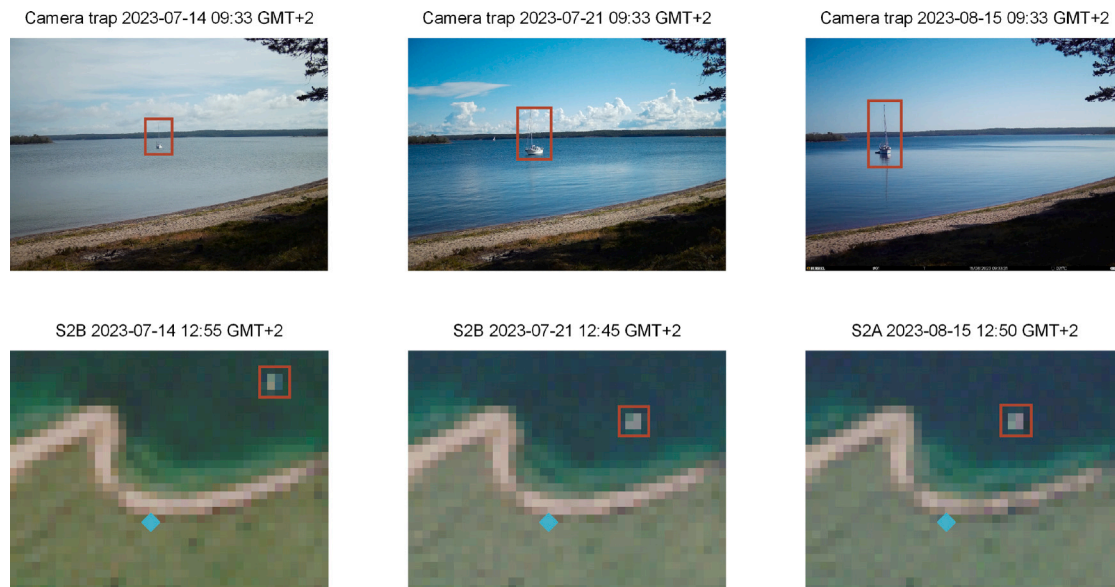


Fig. 3. Pleasure crafts captured both in camera trap in Sandön island (Archipelago Sea) and the corresponding Sentinel-2 acquisition. Camera trap location is marked with a blue diamond.

The annotated data are provided as GeoPackage files, along with the information of which Sentinel-2 products they correspond to and the instructions on how to convert the data into formats suitable for different object detection frameworks. The dataset is available on the Zenodo portal ([10.5281/zenodo.10046341](https://doi.org/10.5281/zenodo.10046341)).

3. Methods

3.1. Object detection models

Object detection models can be divided into three different types of approaches. Two-stage detectors, such as R-CNN (Girshick et al., 2014), Faster R-CNN (Ren et al., 2017) and Mask R-CNN (He et al., 2020) have a separate region proposal step that filters the input image for possible regions that contain an object. Object detection and classification is then performed for these regions only. On the other hand, single-stage object detectors, such as YOLO (short for *You Only Look Once*) (Redmon et al., 2016) do not have a separate region proposal network, but instead use the full image to predict the bounding boxes. For instance, YOLO-based models divide the input image into a $S \times S$ grid or multiple grids, and each grid cell is responsible for detecting objects within it. The third approach is used by transformer-based methods, such as DETR (Carion et al., 2020). These approaches use a CNN backbone to extract features from the input image, which is then fed into a transformer encoder. Typically, single-stage detectors have the lowest performance, but they are also the fastest to both train and to generate predictions.

Since its first release in 2015, YOLO has had several updates and improvements (e.g. Redmon and Farhadi, 2016, 2018; Bochkovskiy et al., 2020; Jocher et al., 2021) by multiple different authors. Since YOLOv3, the models have implemented a multi-scale head to improve the detection of small objects, as well as produce more accurate bounding boxes (Redmon and Farhadi, 2018). The multi-scale head outputs the predictions for three different sized grids (in the case of YOLOv3, 13×13 , 26×26 and 52×52 , which correspond to 32-, 16- and 8-pixel grid cells). The predictions from each scale are then combined using the Non-Maximum Suppression (NMS) algorithm, which selects the best bounding boxes among all the overlapping ones. Since YOLOX (Ge et al., 2021), all later YOLO models have used an anchor-free detection process. Since YOLOv2, the models had relied on anchor boxes (predefined hyperparameters specifying the sizes and scales of the

objects) on detecting the objects, but these had the limitation of being less generalised outside of the training data. YOLOX also introduced the decoupled head, where there were separate convolutional layers responsible for the regression task (location of the bounding box) and classification (identifying the object inside the bounding box) instead of both tasks sharing the same layers and thus same features (Terven et al., 2023).

As the field of object detection and computer vision evolves rapidly, there have been multiple updated iterations of the YOLO model family during the last couple of years. First, YOLOv8 by Ultralytics (Jocher et al., 2023), whose main updates of compared to YOLOv5, the previous iteration developed by the same team, were the adoption of anchor-free detection, decoupled head, and improvements on the convolutional layers of the model backbone. YOLOv8 uses CSPDarknet53 (Wang et al., 2020; Bochkovskiy et al., 2020) as the backbone network to extract features from the input image. Next, YOLOv9 (Wang et al., 2024b) introduced the concepts of programmable gradient information (PGI) and generalised efficient layer aggregation network (GELAN). The purpose of PGI was to improve the problems caused by deep networks, namely the information bottlenecks, whereas GELAN is a new network architecture used as the backbone network. YOLOv10 (Wang et al., 2024a) proposed an NMS-free training and inference, as well as more streamlined backbone architecture. The advantage of NMS-free models is more efficient inference, as well as having fewer expert-defined hyperparameters. The latest iteration of YOLO (at the time of conducting this study) was YOLO11, again developed by Ultralytics. Compared to YOLOv8, YOLO11 has an improved backbone and uses less parameters, while having better benchmark performance (Jocher et al., 2023).

The models from the YOLO-family have been used for various object detection tasks in multiple domains, including ship detection tasks (e.g. Hass and Jökar Arsanjani, 2020; Jiang et al., 2021; Xu et al., 2022; Büyükkarber et al., 2023) with good results. In this work, we tested how four of the most recent iterations (at the time of conducting this study) of the YOLO model family, YOLOv8, YOLOv9, YOLOv10 and YOLO11, perform in the task in hand. We did not make any adjustments to the internal model architecture or layer structure, but only fine-tuned or trained the models from scratch with our own data. Implementations for all these models are available on ultralytics package, so testing and comparing them with mostly identical training configurations was straightforward.

All models were trained for a maximum of 300 epochs so that the training was stopped if the validation metrics did not improve for 30

epochs. The model checkpoint with the best validation score was saved together with the last model. We used the Adam optimiser (Kingma and Ba, 2017) with an initial learning rate of 0.001 and linear learning rate scheduling with a factor of 0.01. The batch size was optimised for each model to be as large as the GPU memory allowed. The 320×320 pixel image chips were upsampled to 640×640 pixels by the model in order to improve the detection rate of the smallest targets, as it increased their size from 2×2 pixels to 4×4 pixels. During training, the input images were augmented with rotations, flips, random scaling, colour balance adjustments, as well as with mosaic augmentation (Bochkovskiy et al., 2020), in which input image is constructed from multiple images from the dataset. All models were trained using a single NVIDIA V100 GPU with 32 GB of GPU memory, hosted on computing nodes of the Puhti supercomputer by CSC — IT Center for Science, Finland. We used python version 3.12.8, Pytorch version 2.6.0 and ultralytics version 8.3.74 in this work.

During testing, we used our models to generate the predictions for the full mosaics and evaluated the results based on the full images. A common approach with processing large images is to process the image into smaller chips using a sliding window approach with a small amount of overlap between chips to reduce edge artefacts. We used SAHI (Akyon et al., 2022) Python package with an overlap ratio of 0.2 to handle the chip generation with overlap and merging predictions in overlapping areas. Finally, we georeferenced the predicted bounding boxes to the CRS of the original image. All predictions were done without using test-time augmentation.

3.2. Evaluation

The object detection performance was evaluated using standard object detection metrics: precision, recall, F1-score (the harmonic mean of precision and recall), mean Average Precision with an Intersection-over-Union (IoU) threshold of 0.5 (mAP50), and mAP@[.5 : .95]. IoU measures the ratio between the area of overlap and the union of two bounding boxes, and all detection with IoU larger than a threshold of, e.g., 0.5 with a ground truth box were defined as True Positives (TP). Detections without corresponding ground truths were defined as False Positives (FP), and ground truth masks without corresponding TP were defined as False Negatives (FN). Precision measures how many of the detected vessels were relevant (ratio of TP and all detected boxes) and recall measures how much of the annotated ships were detected (ratio of TP and all ground truths). Average precision (AP) is the area under the Precision–Recall curve, defined as

$$AP = \int_{R=0}^1 P(R) dR$$

where P and R are the precision and recall values on the Precision–Recall curve, using some IoU threshold. mAP50 is computed using IoU threshold of 0.5, and mAP@[.5 : .95] is the average of AP with IoU thresholds between 0.5 and 0.95. While object detection algorithms are typically also benchmarked for different sizes of objects (small, medium and large), we did not use mAP metrics across different scales to evaluate our results, as almost all our annotations are considered to be small (less than 32×32 pixels) by COCO metric definitions. Validation scores were computed based on all detections, but test results were pre-filtered using the confidence threshold with the highest F1-score for each model. Metrics were evaluated using the implementations in the ultralytics package. The primary metrics for determining which models were the best performing were F1-score and mAP50, as F1-score balances both precision and recall, and as our targets were only a few pixels in size, estimating object detection accuracy with an IoU threshold of 0.5 was sufficient for our purposes.

3.3. Post-processing

As the mosaics contained several stationary objects that are easily confused with marine vessels, we also quality-controlled the predictions further before evaluating the full tiles. First, all predicted polygons whose centroid points were not located in water were discarded. The water mask used here contained the layers for lakes, sea and rivers (original material: the Topographic database by the National Land Survey of Finland). Then, all predictions whose centroid points were located in areas with above-water rocks were discarded, using the data from the Topographic database. We also discarded all predictions that contained a large above-water rock, a lighthouse or a sector light within the bounding box (data from the Finnish Transport Infrastructure Agency). We also discarded all predictions that had either of its edges longer than 750 metres, as these objects are too large to represent a marine vessel in the Baltic Sea, even when including the most visible part of the vessel's wake. For fish farms, we created a layer based on manual interpretation, as there was no available location data for them. For the test results, we compared how this affected the object detection metrics, as well as how much of each error type were present in the results.

4. Results

In this section, we present the results for the trained YOLO models and select the best performing model for further use. First, we present the overall validation and test set results and compare whether using L1C-TCI or Tarkka colour correction affects the results, as well as whether training the model from scratch or fine-tuning from pretrained models produces better validation and test results. For test data, we reported the numbers of true positives, false positives and false negatives in addition to the validation metrics. Second, we compared the test set results acquired with models with the same base architecture, but trained with different types of data, and evaluated what differences the results acquired from the data processed differently have. Finally, as a demonstration of utilising the proposed workflow to monitor maritime traffic over longer time periods, we used the model with the highest test set F1-score to generate a marine traffic map for two boating hotspots on the Finnish coast: the Archipelago Sea and the Gulf of Finland.

4.1. Ship detection evaluation

In total, we trained 84 different models using fixed random seed and identical hyperparameters aside from batch size, which was set to be such that 60% of GPU memory was used. This ensured that the same model configurations were trained with identical conditions (e.g., same annotations and image locations at the same order).

Based on validation results, fine-tuned models achieved higher F1 and mAP50 scores than the models trained from scratch, no matter which input data were used. The differences were fairly minor, as the F1 scores for the top 50% of the trained models ranged between 0.827 and 0.795, and mAP50 was between 0.871 and 0.833. The clear outliers among the trained models were the two yolov10x models trained from scratch, as they achieved F1 and mAP50 scores of less than 0.6, being the only ones to have an F1 score less than 0.73 or mAP50 less than 0.75. As for the input data, models trained with Tarkka data typically achieved slightly higher metrics than models using L1C-TCI data. However, as the reference data were created based on the Tarkka data, annotations may have a bias towards those data. The average results for each model family, grouped by data types and whether the models were pretrained or not, are shown on Table 3. Based on the validation results, the individual models with the highest F1-scores were yolo11m trained with Tarkka data (F1 0.827, mAP50 0.867), yolo11s trained with Tarkka data (F1 0.827, mAP50 0.871), yolo11s trained with L1C-TCI data (F1 0.825, mAP50 0.853) and yolov9m trained with Tarkka

Table 3

Average validation results for each model family, grouped by data and training type.

Model family	Data type	Train type	Precision	Recall	F1	mAP50	mAP
yolov8	Tarkka	Pretrained	0.829	0.809	0.819	0.855	0.414
		Scratch	0.816	0.758	0.789	0.820	0.382
	L1C-TCI	Pretrained	0.841	0.783	0.811	0.839	0.387
		Scratch	0.823	0.762	0.791	0.815	0.368
yolov9	Tarkka	Pretrained	0.838	0.792	0.814	0.855	0.415
		Scratch	0.832	0.774	0.802	0.835	0.390
	L1C-TCI	Pretrained	0.836	0.797	0.816	0.845	0.395
		Scratch	0.816	0.771	0.792	0.827	0.381
yolov10	Tarkka	Pretrained	0.807	0.749	0.776	0.827	0.403
		Scratch	0.764	0.704	0.733	0.774	0.366
	L1C-TCI	Pretrained	0.786	0.723	0.753	0.801	0.379
		Scratch	0.757	0.695	0.725	0.761	0.352
yolo11	Tarkka	Pretrained	0.835	0.780	0.806	0.853	0.421
		Scratch	0.826	0.778	0.802	0.836	0.396
	L1C-TCI	Pretrained	0.832	0.779	0.805	0.838	0.394
		Scratch	0.798	0.744	0.770	0.795	0.359

data (F1 0.824, mAP50 0.860). Each of these models was finetuned from pretrained weights.

However, for the test dataset, the models trained with L1C-TCI data acquired slightly better F1-scores than the models trained with the Tarkka data, which is opposite of the validation results. Typically, the models detected fewer potential vessels from the L1C-TCI images than from the Tarkka images, and while the models using different data had similar numbers of false negatives, L1C-TCI had fewer false positives. Overall, the test set results before post-processing (Table 4) were lower than validation results (except for recall), and all models had a large number of false positives.

The results for the test dataset improved significantly after removing the stationary targets (Table 5), as the mean precision score for all models increased from 0.726 to 0.861, F1-score from 0.759 to 0.825 and mAP50 from 0.745 to 0.874. As this post-processing decreases the number of false positives, the recall scores remained the same. The most common false positives that were cleaned at this stage were detections not on water, and detections that contained an above water rock. On tile 34VEN, fish farms were also a fairly common false positive target. Of the individual models, the highest F1-scores after post-processing were achieved by yolo11s trained with L1C-TCI data (F1 0.863, mAP50 0.888), yolov8x trained with L1C-TCI data (F1 0.860, mAP50 0.898) and yolov8l trained with L1C-TCI data (F1 0.859, mAP50 0.912), all fine-tuned from pretrained weights. However, most of the models achieved good results, as half of the models had an F1-score of more than 0.830 or mAP50 of more than 0.880. There were several models that achieved either precision or mAP50 of more than 0.9 but had considerably lower recall. Based on these results, we selected yolo11s utilising L1C-TCI data as the model to produce the yearly marine traffic data. This model checkpoint is available on <https://huggingface.co/mayrajeo/marine-vessel-detection-yolo>.

4.2. Comparison of individual scenes in the test dataset

In this section, we inspect the individual scenes from the test dataset further, and examine which differences the models trained with different data had. For these analyses, we used yolo11s fine-tuned from pretrained weights, and compare the results acquired with both data processing types. Both yolo11s models had similar validation results, but the model utilising L1C-TCI data had higher test set metrics after post-processing (F1: L1C-TCI 0.863 vs. Tarkka 0.823; mAP50: L1C-TCI 0.888 vs. Tarkka 0.831).

The test dataset consisted of two separate locations, of which 34VEN had four scenes and 34VER had three. The scene-wise results for both models are shown on Table 6. Overall, both models achieved good

results for all scenes except for 2022-06-19, and the main difference between them was the L1C-TCI model detecting fewer false positives in all scenes. One of the scenes in 34VEN (2022-06-19) contained rougher sea conditions compared to the other images, in order to evaluate the models' performance in these types of conditions. For both models, this specific scene clearly had the worst results, as the F1-score for the Tarkka model was only 0.487, and the L1C-TCI model, while performing better, still had an F1-score of only 0.610. The main difference between the models, considering this image, was the Tarkka model incorrectly detecting waves in the rough sea conditions as vessels, while the false positives for the L1C-TCI model were mostly above-water rocks that were not included in the data used for post-processing.

4.3. Marine traffic in the Archipelago Sea and the Gulf of Finland during summer 2022

To demonstrate the use of models for quantifying maritime traffic in Finnish marine areas, we applied the top-performing model, yolo11s trained with L1C-TCI data, to process all available products for the Archipelago Sea and the Gulf of Finland, from 1 May to 30 September 2022. Fig. 4 illustrates these areas, along with the Sentinel-2 tile coverage. We used a confidence threshold of 0.286 for the predictions, determined by the highest validation F1-score of the used model.

The Archipelago Sea area is covered by 4 Sentinel-2 tiles (34VDM, 34VDN, 34VEM, 34VEN). Our test area contains only the Finnish territorial waters instead of the full area covered by the four Sentinel-2 tiles, as almost all recreational boating occurs in that area. Second, limiting the area this way reduces the amount of open sea, where the used model had some challenges due to rougher sea conditions, evident in one of the validation images. Acquisition times for these areas were between 12:30 PM and 1:30 PM local time. We filtered the data to only contain relative orbits (the track identifiers of the satellite overpasses) 122 and 022, as they both cover almost the full area of the four tiles. Using multiple relative orbits also decreased the temporal resolution to two or three days. This resulted in 61 individual products for each of the analysed tiles. Out of these acquisitions, we discarded 23 from further analyses. One tile was discarded due to rough sea conditions, in which the model incorrectly detected waves, glint or other rough sea conditions in the open sea as marine vessels, resulting in large number of FPs. The other 22 discarded tiles were covered by clouds so that the area of interest was almost completely obscured. Our final data for the Archipelago Sea analyses contained 38 acquisitions of sufficient quality: nine from May, ten from June, six from July, seven from August and six from September. Of these acquisitions, 32 were from weekdays (Monday–Friday) and six were from weekends.

In the Archipelago Sea, the most active days were Saturday 13 August 2022 (2104 detected vessels), Friday 24 June 2022 (1462 detected vessels) and Tuesday 19 July 2022 (1055 detected vessels). On average, July was the most active boating month, with 765 detected vessels per acquisition, followed by August (650) and June (588). As expected, we observed more traffic during weekends than on weekdays: on average, we detected 616 vessels per day during weekends and 417 vessels per acquisition on weekdays. Most of the detected traffic was focused on marine fairways and near the main harbours of larger cities (Turku, Naantali, Uusikaupunki, Rauma and Mariehamn). Fig. 5 shows the detected vessels in a 1 × 1 km grid, aggregated by the month and the overall detections. Overall, it seems that maritime traffic concentrates in the central parts of the Archipelago Sea and is most active during July. We can observe increasing traffic activity entering the Archipelago Sea from the Gulf of Finland in June and July (the busy boating fairway in the southeastern corner of Fig. 5). During other months, traffic shifts to the inner archipelago and closer to the coast, likely due to the end of the holiday season.

The Gulf of Finland is covered by two Sentinel-2 tiles (35VMG and 35VLG), and using relative orbits 079 and 036 provided us with mostly complete coverage of the two tiles. The acquisition times for these

Table 4

Average non-filtered test set results for each model family, grouped by data and training types.

Model family	Data type	Train type	Precision	Recall	F1	mAP50	mAP	TP	FP	FN
yolov8	Tarkka	Pretrained	0.692	0.836	0.757	0.732	0.371	1416.4	638.2	277.6
		Scratch	0.707	0.790	0.745	0.728	0.368	1338.4	559.8	355.6
	L1C-TCI	Pretrained	0.737	0.838	0.784	0.759	0.371	1418.8	508.0	275.2
		Scratch	0.700	0.818	0.754	0.716	0.344	1386.4	601.8	307.6
yolov9	Tarkka	Pretrained	0.735	0.820	0.774	0.764	0.391	1389.0	506.8	305.0
		Scratch	0.708	0.815	0.757	0.736	0.367	1380.0	576.8	314.0
	L1C-TCI	Pretrained	0.721	0.837	0.775	0.746	0.366	1418.6	550.6	275.4
		Scratch	0.762	0.798	0.779	0.776	0.370	1351.6	429.4	342.4
yolov10	Tarkka	Pretrained	0.771	0.774	0.771	0.788	0.398	1310.7	395.3	383.3
		Scratch	0.682	0.750	0.714	0.678	0.328	1271.3	590.5	422.7
	L1C-TCI	Pretrained	0.798	0.767	0.780	0.805	0.389	1298.8	337.2	395.2
		Scratch	0.737	0.754	0.744	0.736	0.344	1277.5	459.5	416.5
yolo11	Tarkka	Pretrained	0.733	0.800	0.760	0.765	0.386	1354.8	522.0	339.2
		Scratch	0.686	0.818	0.744	0.710	0.356	1385.4	646.8	308.6
	L1C-TCI	Pretrained	0.754	0.823	0.787	0.774	0.382	1393.8	456.8	300.2
		Scratch	0.678	0.786	0.726	0.694	0.329	1331.8	644.2	362.2

Table 5

Average test set results for each model family after post-processing steps, grouped by data and training types.

Model family	Data type	Train type	Precision	Recall	F1	mAP50	mAP	TP	FP	FN
yolov8	Tarkka	Pretrained	0.834	0.831	0.832	0.859	0.437	1408.0	283.4	286.0
		Scratch	0.874	0.785	0.827	0.887	0.449	1330.2	192.2	363.8
	L1C-TCI	Pretrained	0.871	0.833	0.851	0.886	0.434	1411.2	211.2	282.8
		Scratch	0.852	0.813	0.831	0.867	0.418	1377.2	242.6	316.8
yolov9	Tarkka	Pretrained	0.862	0.815	0.837	0.881	0.452	1381.0	224.0	313.0
		Scratch	0.865	0.810	0.836	0.882	0.441	1372.0	215.0	322.0
	L1C-TCI	Pretrained	0.846	0.833	0.839	0.866	0.426	1411.6	258.0	282.4
		Scratch	0.868	0.792	0.828	0.879	0.420	1342.4	205.0	351.6
yolov10	Tarkka	Pretrained	0.884	0.769	0.822	0.894	0.452	1302.8	172.8	391.2
		Scratch	0.841	0.746	0.790	0.836	0.404	1264.2	234.2	429.8
	L1C-TCI	Pretrained	0.892	0.763	0.821	0.900	0.435	1292.3	160.0	401.7
		Scratch	0.863	0.750	0.801	0.863	0.404	1269.8	202.0	424.1
yolo11	Tarkka	Pretrained	0.853	0.795	0.821	0.872	0.442	1346.0	238.2	348.0
		Scratch	0.847	0.813	0.828	0.867	0.436	1377.4	254.6	316.6
	L1C-TCI	Pretrained	0.878	0.818	0.847	0.891	0.440	1386.4	194.0	307.6
		Scratch	0.842	0.781	0.809	0.853	0.405	1322.2	252.4	371.8

Table 6

Comparison of individual scene results between fine-tuned yolo11s models trained with different input data.

Data type	Scene	Date	Precision	Recall	F1	mAP50	mAP	TP	FP	FN
Tarkka	34VEN	2021-07-14	0.858	0.86	0.859	0.884	0.438	387	64	63
		2022-06-19	0.337	0.879	0.487	0.345	0.151	58	114	8
		2022-06-24	0.850	0.828	0.839	0.853	0.436	351	62	73
		2022-08-13	0.885	0.905	0.895	0.914	0.502	361	47	38
		2022-06-17	0.726	0.638	0.679	0.706	0.355	53	20	30
		2022-07-12	0.851	0.777	0.813	0.888	0.443	143	25	41
		2022-08-26	0.815	0.852	0.833	0.837	0.442	75	17	13
	L1C-TCI	2021-07-14	0.930	0.862	0.895	0.940	0.464	388	29	62
		2022-06-19	0.486	0.818	0.610	0.490	0.232	54	57	12
		2022-06-24	0.882	0.830	0.855	0.891	0.442	352	47	72
		2022-08-13	0.931	0.917	0.924	0.945	0.481	366	27	33
		2022-06-17	0.826	0.687	0.75	0.854	0.429	57	12	26
		2022-07-12	0.910	0.826	0.866	0.933	0.449	152	15	32
		2022-08-26	0.829	0.773	0.800	0.881	0.456	68	14	20

data were between 1:30 PM and 3:00 PM local time. Altogether, 45 out of 61 products were deemed suitable for further analyses: seven from May, ten from June, eleven from July, ten from August and seven from September. All discarded products were due to cloud coverage. From weekdays, there were 31 products and 14 from weekends. Like in the Archipelago Sea, the boat traffic was more active during weekends than weekdays, as the average number of vessels during weekends was 575 compared to 244 during weekdays. The highest boating activities were detected during 26 June 2022, 21 July 2022 and 27 August 2022

with 1486, 1056 and 953 detected vessels, respectively. August had the highest detected daily traffic, with 475 detections per acquisition, followed by July (408) and June (402).

Most of the detected traffic was focused near Helsinki and Espoo, and overall, the traffic was focused close to the coastline, as seen in Fig. 6. Compared to the Archipelago Sea, the locations for the detected vessels were more consistent between months, and the traffic is concentrated close to the coastline. There were also a few vessel hotspots away from the coast, which correspond to the anchoring areas

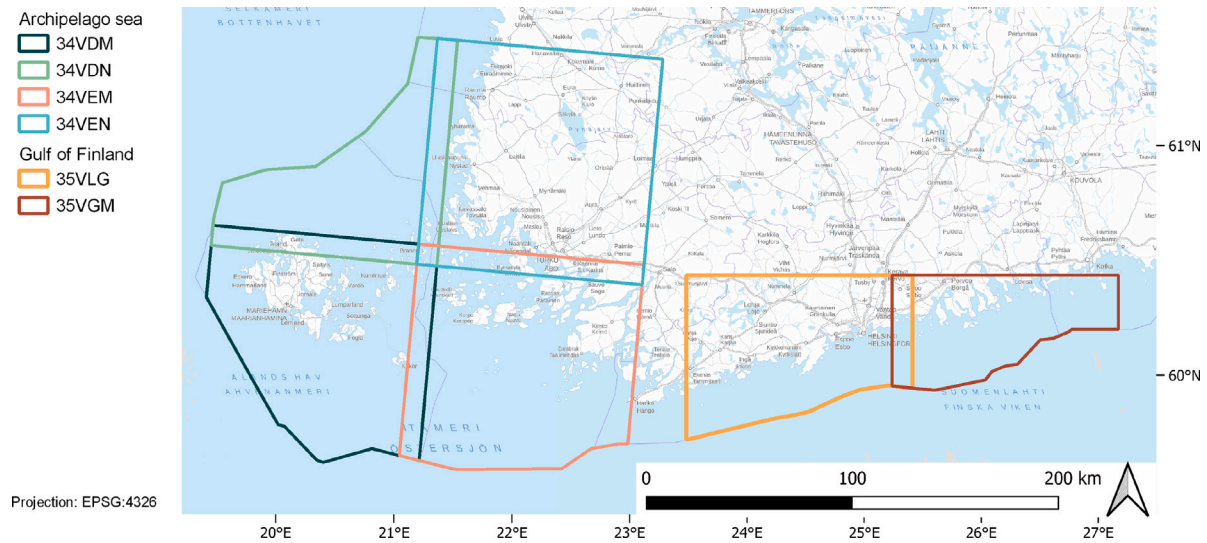


Fig. 4. Test areas and identifiers of Sentinel-2 tiles covering them. Background map: The National Land Survey Finland.

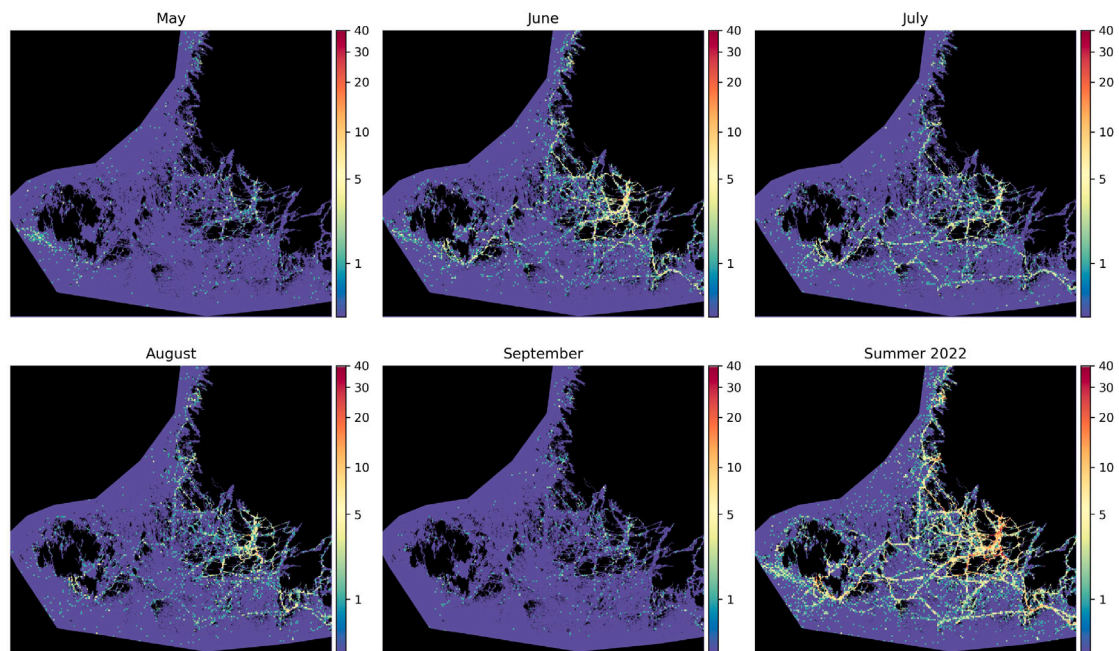


Fig. 5. Monthly and yearly marine traffic heat maps for the Archipelago Sea. The values signify the number of detected vessels in a 1×1 km grid cell.

for the cargo and tanker ships in the Gulf of Finland. As for the monthly differences, in June and August, the traffic is more concentrated near the Helsinki metropolitan area, but in July, there is a clear increase in the numbers of detected vessels in the western parts of the area. One possible explanation for this is that the maritime traffic shifts from the Helsinki metropolitan area to the Archipelago Sea in July, due to the main holiday driving the recreational activities (Strandell et al., 2020; Paulus et al., 2024). This could also explain why the eastern part of the area had clearly less traffic compared to the western parts.

5. Discussion

In this study, we evaluated the suitability of the YOLO model family object detection models for detecting and monitoring maritime traffic from optical satellite imagery. Moreover, we evaluated whether colour adjustment developed specifically for the area under study was beneficial compared to L1C-TCI images available in the Sentinel-2

products. The proposed method of utilising object detection models from the YOLO model family for marine vessel detection achieved good results, with overall precision and recall being both higher than 0.8, and mAP50 being around 0.85 on average. According to our validation and test results, finetuning the models from pretrained checkpoints produced better results than training the models from scratch. Especially for the largest models (yolov8x, yolov9e, yolov10x and yolo11x), training the models from scratch proved to be challenging, as the batch sizes for these models had to be low due to memory constraints. As for how the specialised colour adjustment method affected the results, the results indicated it did not have a substantial beneficial effect for the models. Even though it was helpful for human experts to interpret the images corrected with these methods, the models using these methods did not significantly outperform the models using L1C-TCI data in validation metrics, and, in fact, L1C-TCI data produced better metrics for the test dataset. These results show that this type of approach is viable for detecting marine vessels, including pleasure crafts from satellite imagery, even with data of 10-metre pixel resolution.

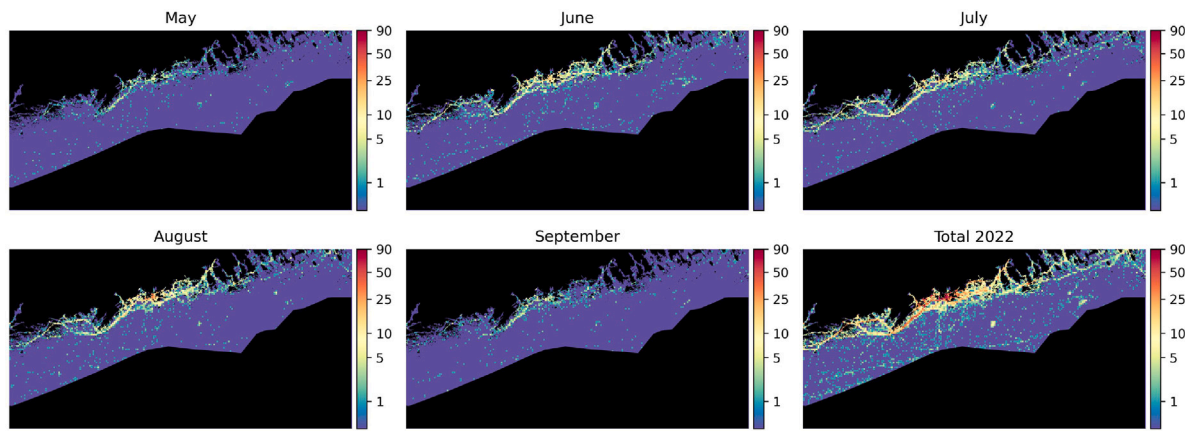


Fig. 6. Monthly marine traffic heat maps for the Finnish waters of the Gulf of Finland. The values signify the number of detected vessels in a 1×1 km grid cell.

Most of the studies focused on ship detection are either focused on larger, commercial ships (e.g. Paolo et al., 2024), or utilise VHR data for detecting the targets (e.g. Smigaj et al., 2023). As for the studies focusing on small vessel detection, it is worth noting that the definition of a small marine vessel varies between different studies, and it can mean vessels smaller than 300 tonnes, or small boats shorter than 5 metres (Zucchetta et al., 2025). Aiello et al. (2019) evaluated the detection rates of different-sized ships from high-resolution (less than 2-metre spatial resolution), medium-resolution (Sentinel-2) and SAR imagery, and while most vessels with a length of less than 30 metres were detected from HR images, most of the same-sized targets remained undetected from Sentinel-2 images. Ciocarlan and Stoian (2021) defined small ships as a target with an area less than 2500 m² (i.e. 5×5 pixels in Sentinel-2 data), and the detection rate for small ships was lower compared to larger targets. A common challenge in detecting small ships is that they are mixed with rough sea conditions or other ship-like targets (i.e., small islands or coastal rock formations) Aiello et al. (2019) and Zucchetta et al. (2025). We also observed similar challenges with our data, especially evident in one of the validation images (34VEN dated 19 June 2022), more apparent with Tarkka images.

Post-processing the predictions with external databases to filter stationary non-vessel targets proved to be useful, as a large number of objects, such as sector lights or wind turbines, look like marine vessels in Sentinel-2 imagery. This process increased the precision of the models by more than 0.1 and eliminated around half of the false-positive detections. Unfortunately, the openly available external data used did not cover all cases, such as fish farms, which meant that the data for them had to be manually drawn based on the satellite images. However, even if there are no available data for stationary objects, manually creating the data can be used with multiple acquisitions, and we consider these types of external data essential for improving the quality of the predictions.

While observing the predicted seasonal boat traffic for the Archipelago Sea and the Gulf of Finland, we observed a few acquisitions with a large numbers of false positives. These situations occurred for images with rougher sea conditions in the more open areas, and the model detected wakes or glint as vessels. This kind of surface clutter typically corresponds to high wind speeds, and are a common problem in marine vessel detection, as they cause false detections (Dekker et al., 2013; Kanjir et al., 2018). In some cases, detecting small vessels from rough seas is almost impossible even for human experts. Even though our reference dataset contained these types of situations, it did not completely succeed in making the models handle them, as seen with one of the test images. While manually selecting the useable products was straightforward for our demonstration cases, it is not as trivial for, e.g., analysing the whole Finnish Coast over multiple years. For instance, the number of products to evaluate can be hundreds of

images even for a single year, making it a time-consuming task. Still, selecting the useable products or filtering obviously faulty products is far less cumbersome than manually delineating marine vessels from the images. Also, it is worth noting that using only data from a single year limits the information that can be derived from the results. In the future, we aim to utilise all available data for the analyses to uncover the trends between years, as well as more accurately detect the hotspots.

Because the resolution of Sentinel-2 imagery is rather coarse for detecting small-sized vessels, such as pleasure crafts, detecting these objects can be difficult even for expert annotators. The presented method does not enable classifying the detected vessels into different boat categories aside from categorising by the size of the bounding box, and thus the detections contain some amount of commercial marine traffic. Of course, the commercial traffic using AIS can be identified by comparing the AIS data and detections with matching time steps, but AIS data is typically not openly available. Moving objects can be detected based on their wakes, but stationary objects require comparison of multiple acquisitions to be spotted by expert annotators. Moreover, detecting vessels in the harbours or attached to the shore is not realistic with these data, as these targets blend into their surroundings. Detecting these vessels, as well as more accurately classifying them based on vessel categories, would require higher resolution imagery, and there are already several works that show excellent performance on VHR imagery (e.g., Büyükkanber et al., 2023 and Kong et al., 2023). While commercial satellites can already provide these kinds of data, using them for long-term, large-scale monitoring is both computationally and financially expensive. However, future generations of openly available satellite data might provide data with sufficient resolution to more accurately distinguish between various vessel types. The anticipated improvements in the resolution and spectral constellation of Sentinel-2 Next Generation (S2NG) instruments will provide a prominent source for improving the methodology.

In this study, we restricted the analysed dataset to optical satellite data quantised into RGB images. The applied instrument, Sentinel-2 MSI, has ten additional spectral bands, but only one of those bands has 10-metre resolution (B08, near-infrared). Other bands have either 20- or 60-metre resolution, which limits their usefulness for detecting small objects. Utilising only three bands also makes it more straightforward to use existing object detection frameworks, as most of the computer vision research, and thus the pretrained models, are focused on traditional RGB data. However, recent research has focused on utilising models pretrained on RGB data with multispectral input, usually by copying the weights for the first layers (e.g. Ciocarlan and Stoian, 2021). Nevertheless, the main downside of using optical images is the inability to see through clouds. We had to discard around a third of the images from 2022 while classifying the marine traffic in the test areas due to almost complete cloud coverage, and even the remaining

images had typically at least partial cloud coverage. Using SAR data would help with the cloud coverage, and it has been successfully used for this task (e.g., Ding et al., 2023; Kong et al., 2023; Ruciński et al., 2023). The overall workflow to implement this type of methodology for SAR imagery is slightly more laborious than optical data with respect to, e.g., pre-processing steps and manual annotation of marine vessels, especially in the case of smaller recreational boats. Moreover, at the time of writing, only one Sentinel-1 satellite was operational, meaning that the temporal gap between images is larger compared to Sentinel-2 data. Also, the acquisition times for Sentinel-1 in the Finnish coast is not optimal for monitoring recreational marine traffic, as the descending image is captured at around 7 AM local time, and the ascending image at around 7 PM local time. Nevertheless, utilising SAR imagery would also enable detecting the vessels on cloudy days, and testing the presented approach with it, or implementing a workflow utilising both SAR and optical data is an interesting topic for a follow-up study.

To quantify the impacts of recreational boating on marine biodiversity, the next logical step is to link the potential pressures resulting from vessel activity to underwater biological inventory data (e.g. Forsblom et al., 2024). This can, however, be a daunting task, as species composition varies in response to environmental parameters, such as salinity, the level of eutrophication, turbidity, exposure and light availability. Disentangling the effect of boating from these other parameters would require extensive pre- and post-season biological surveys in areas where impacts can be anticipated (e.g., marinas). As such data is currently unavailable, an alternative approach is to approximate potential impacts using EO data. Object detection models, such as the one we used here, can first be employed to quantify vessel activities (i.e., their visiting frequency) in marinas or along boating lanes, where impacts on underwater nature are expectedly high (e.g., enclosed bays, lagoons) (Sagerman et al., 2020). Then, the (increasing) influence of vessel activity on turbidity can be assessed by comparing these locations with areas where vessels are absent. Additionally, if regulative measures are implemented to mitigate the potential impact of boating on marine nature (e.g., speed limits, moving restrictions), object detection models can also be used to monitor the effectiveness of these measures by quantifying changes in vessel activity over time. This way, the presented workflow can both provide crucial information on the pressures exerted by recreational boating and the effectiveness of management measures.

The presented method provides the means to identify areas of intensive maritime traffic, at high temporal and spatial resolution over broad geographical scales. The approach can detect both small leisure-use boats and large commercial vessels from Sentinel-2 imagery. The information acquired can be used to assess the recreational value of coastal areas where people spend time, as well as the potential impacts of boating on marine nature. Such information is useful for the planning of marine protected area networks or restoration locations and for marine spatial planning. Moreover, information can be used to estimate potential, cumulative impacts of human activities on marine nature, and it allows detailed impact assessments of boating activities on SAV in vulnerable locations, such as shallow enclosed bays. To our knowledge, this is the first study to map vessel activity, including recreational boats, at this scale in the northern Baltic Sea. In the future, we aim to test this workflow for other sources of satellite imagery, both higher resolution optical imagery as well as SAR imagery, and to compare our results with AIS data. We aim to build a framework which enables yearly monitoring of marine traffic, including recreational boating on the Finnish coast, and use that information to identify the hotspots of recreational boating over broad geographical scales.

6. Conclusions

In this study, we demonstrated the viability of modern object detection methods for monitoring marine traffic, including recreational boating, from openly available satellite imagery. The models achieved

excellent validation and test performance compared to human annotated data. This indicates that these methods can be used to detect the areas with the most marine traffic and be used further to assess the human pressures and disturbances in shallow areas. It can also be used to evaluate the trends and quantity of boats over seasons and multiple years. We also demonstrated how to utilise external datasets to improve the quality of the detections, as they were used to clean stationary targets from the predictions. As marine traffic on the Finnish coast has not been quantified before at this scale, our results will support ongoing conservation efforts and sustainable use of marine areas. In addition, we made our annotated dataset open source, which can be used for developing further solutions, for example, by utilising them alongside other marine vessel detection datasets.

CRedit authorship contribution statement

Janne Mäyrä: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Data curation. **Elina A. Virtanen:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Ari-Pekka Jokinen:** Writing – review & editing, Writing – original draft, Data curation. **Joni Koskikala:** Writing – review & editing, Writing – original draft. **Sakari Väkevä:** Writing – review & editing, Writing – original draft, Software. **Jenni Attila:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Dataset, analysis examples and models are available on online platforms.

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